Practical 6: Designing Effective and Visually Appealing Dashboards

Aim: To design professional and interactive dashboards in Tableau by applying formatting, colour schemes, customized tooltips, Viz in Tooltip, and visual storytelling techniques for effective business decision-making.

a) Apply worksheet-level and workbook-level formatting

b) Customize number, date, and null value formatting

c) Design effective colour schemes and layouts

d) Create informative tooltips

e) Implement "Viz in Tooltip"

f) Improve readability and user experience

g) Apply best practices for visual storytelling

A.) Aim:- Apply worksheet-level and workbook level formatting.

Step 1: Connect the Dataset

1. Open **Tableau Desktop**.
2. Under **Connect**, click **Microsoft Excel**.
3. Select **Retail_Store_Performance_500.xlsx**.
4. Drag the **RetailData** sheet onto the canvas.
5. Click **Sheet 1**.

Step 2: Create the Required Worksheets

Before starting sub-practicals (A) – (G), create the following worksheets.

Worksheet 1: Sales by Region

1. Rename **Sheet1** to **Sales by Region**.
2. Drag **Region** to the **Columns** shelf.
3. Drag **Sales** to the **Rows** shelf.
4. Click **Show Me** and select **Bar Chart**.
5. Click **Sort Descending**.
6. Drag **Sales** to **Label** on the Marks card.

Worksheet 2: Monthly Sales Trend

1. Create a new worksheet.
2. Rename it **Monthly Sales Trend**.
3. Drag **Order Date** to the **Columns** shelf.
4. Change the date level to **Month (Continuous)**.
5. Drag **Sales** to the **Rows** shelf.

Worksheet 3: Profit by Product Category

1. Create a new worksheet.
2. Rename it **Profit by Category**.
3. Drag **Product Category** to **Rows**.
4. Drag **Profit** to **Columns**.
5. Select **Horizontal Bar Chart**.
6. Sort in descending order.

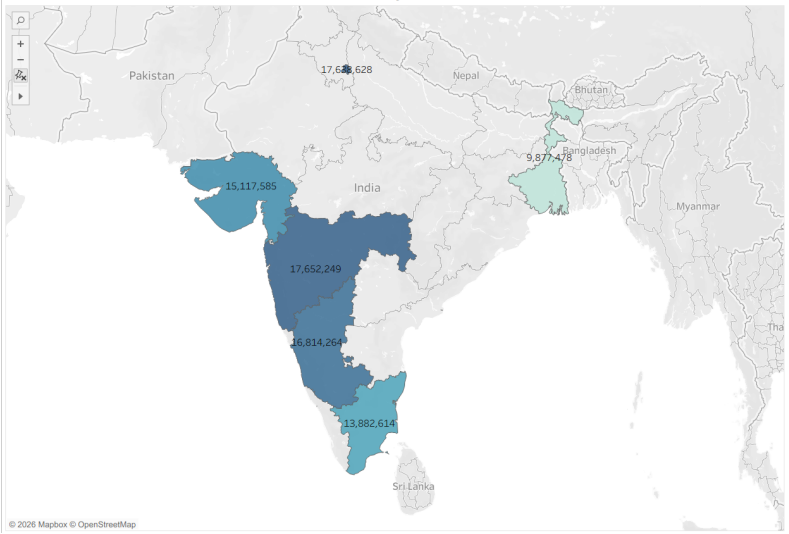
Worksheet 4: Sales by State

1. Create a new worksheet.
2. Rename it **Sales by State**.
3. Double-click **State**.
4. Tableau creates a map automatically.
5. Drag **Sales** to **Color**.
6. Drag **Sales** to **Size**.

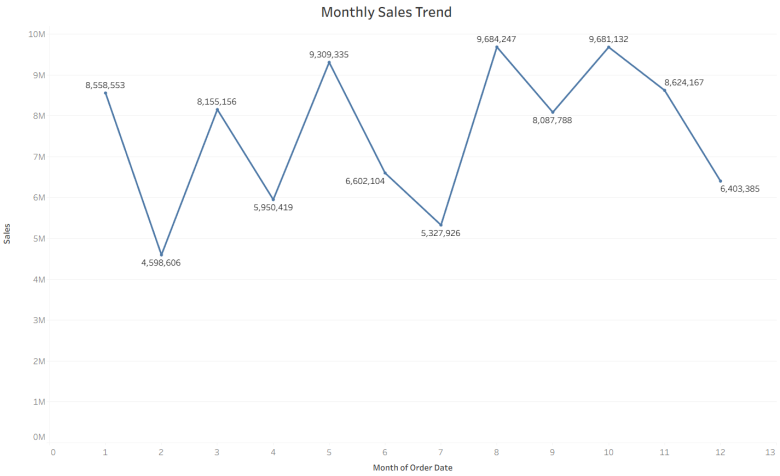
Worksheet 5: Customer Rating Distribution

1. Create a new worksheet.
2. Rename it **Customer Rating Distribution**.
3. Right-click **Customer Rating**.
4. Select **Create → Bins**.
5. Click **OK**.
6. Drag **Customer Rating (bin)** to **Columns**.
7. Drag **Number of Records** to **Rows**.
8. Select **Histogram**.

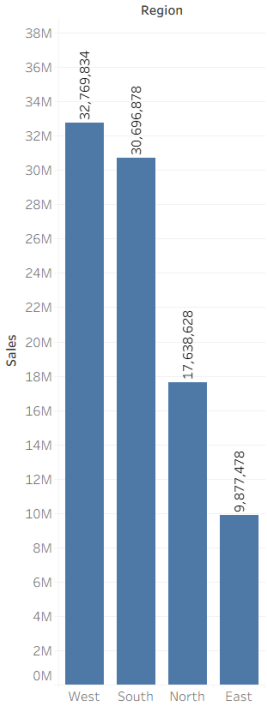
Sales By State



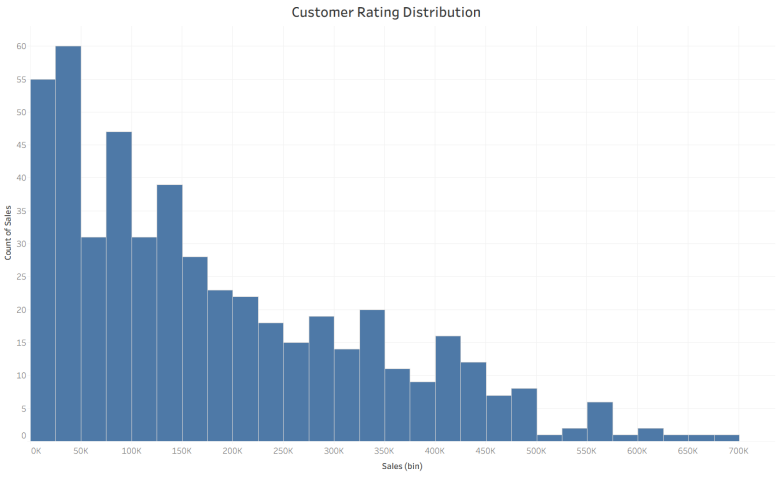
Monthly Sales Trend



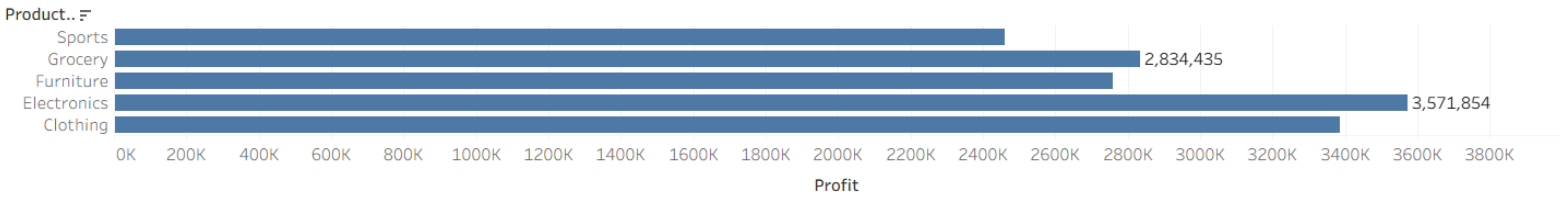
Region



Customer Rating Distribution



Profit By Product



Step 3: Create the Dashboard

1. Click **New Dashboard**.
2. Set **Dashboard Size → Automatic**.
3. Drag all five worksheets onto the dashboard.
4. Arrange them neatly.

After completing these steps, begin the following sub-practicals.

(a) Apply Worksheet-Level and Workbook-Level Formatting

Worksheet-Level Formatting

1. Open the **Sales by Region** worksheet.
2. Double-click the worksheet title and rename it to **Sales by Region**.
3. Click **Format → Font**.
4. Set:
    - Title: 16 pt, Bold
    - Header: 12 pt
    - Pane: 11 pt
5. Click **Format → Borders**.
6. Remove unnecessary row and column dividers.
7. Click **Format → Shading** and keep the worksheet background white.
8. Repeat the same formatting for all worksheets.

Workbook-Level Formatting

1. Open the dashboard.
2. Set **Dashboard Size → Automatic**.
3. Use the same font style throughout the dashboard.
4. Align all charts properly.
5. Maintain equal spacing between charts.
6. Add the dashboard title:
   Retail Store Performance Dashboard

Result:

A professionally formatted dashboard with a consistent appearance.

(b) Customize Number, Date, and Null Value Formatting

Steps:

1. Right-click **Sales**.
2. Select **Default Properties → Number Format**.
3. Choose **Currency (Custom)**.
4. Set decimal places to **0**.
5. Repeat the same steps for **Profit**.
6. Right-click **Discount (%)**.
7. Select **Number Format → Percentage**.
8. Right-click **Order Date**.
9. Select **Default Properties → Date Format**.
10. Choose **MMM YYYY**.
11. If null values are present, click **Format → Special Values**.
12. Replace **Null** with **No Data**.

Expected Output:

Sales and Profit are displayed as currency, Discount is shown as a percentage, dates are displayed in **MMM YYYY** format, and null values appear as **No Data**.

(c) Design Effective Color Schemes and Layouts

Steps:

1. Open the Sales by Region worksheet.
2. On the Marks card, click Color.
3. Choose a suitable color palette (for example, Blue, Green, or Orange).
4. Repeat the same process for all worksheets using consistent colors.
5. Avoid using too many different colors in the dashboard.
6. Open the Dashboard.
7. Arrange all worksheets neatly using Horizontal and Vertical Containers (optional).
8. Keep equal spacing between charts.

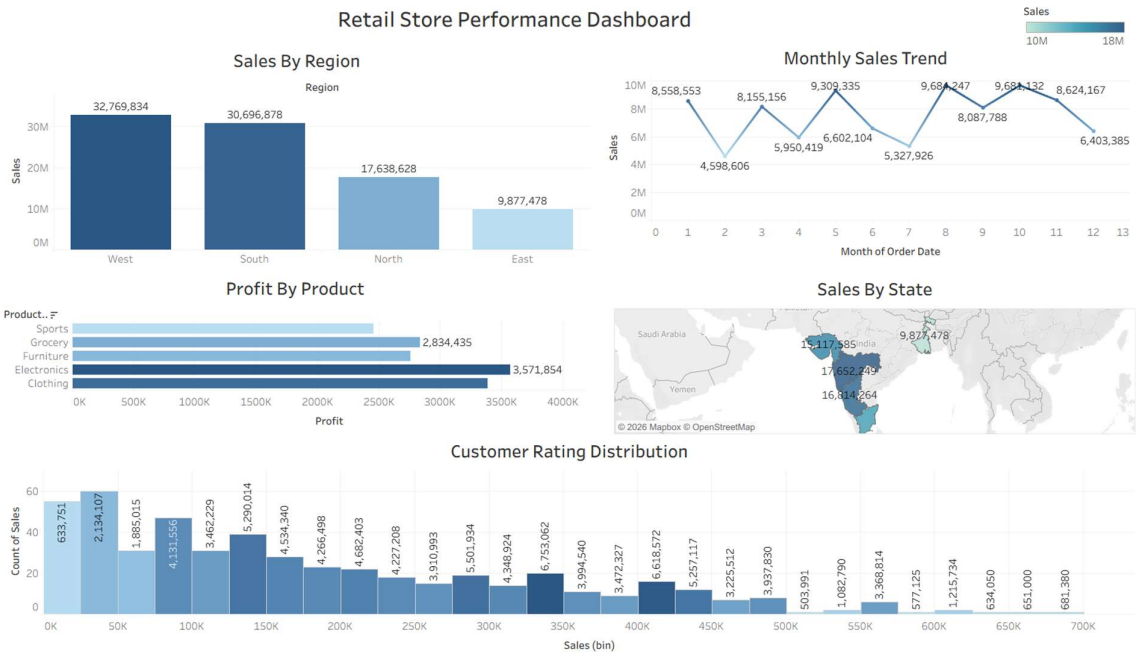
9. Resize charts so that each visualization is clearly visible.
10. Place legends and filters where they do not overlap with charts.
11. Use a white or light-colored dashboard background for better readability.

Expected Output:

The dashboard displays a consistent color theme with properly aligned charts, making the visualization attractive, organized, and easy to understand.

Use Layout Containers (Professional Method)

1. From the left panel, drag a **Horizontal Container** onto the dashboard.
2. Place **Sales by Region** and **Monthly Sales Trend** inside it.
3. Drag a second **Horizontal Container** below the first.
4. Place **Profit by Category** and **Sales by State** inside it.
5. Place **Customer Rating Distribution** below them (or in another container if desired).



(d) Create Informative Tooltips

Theory:

Tooltips provide additional information when the user places the mouse pointer over a mark in a visualization. Informative tooltips help users understand the data without displaying too much information directly on the chart.

Steps:

1. Open the **Sales by Region** worksheet.
2. On the **Marks** card, click **Tooltip**.
3. Delete the default tooltip text if required.
4. Insert the following fields:
    - Region
    - Sales
    - Profit
    - Customer Rating
    - Returned
5. Format the tooltip text with proper labels and spacing.
6. Click **OK**.
7. Repeat similar steps for other worksheets if required.
8. Hover the mouse over any bar to verify the tooltip information.

Example Tooltip

Region : <Region>

Sales : ₹<SUM(Sales)>

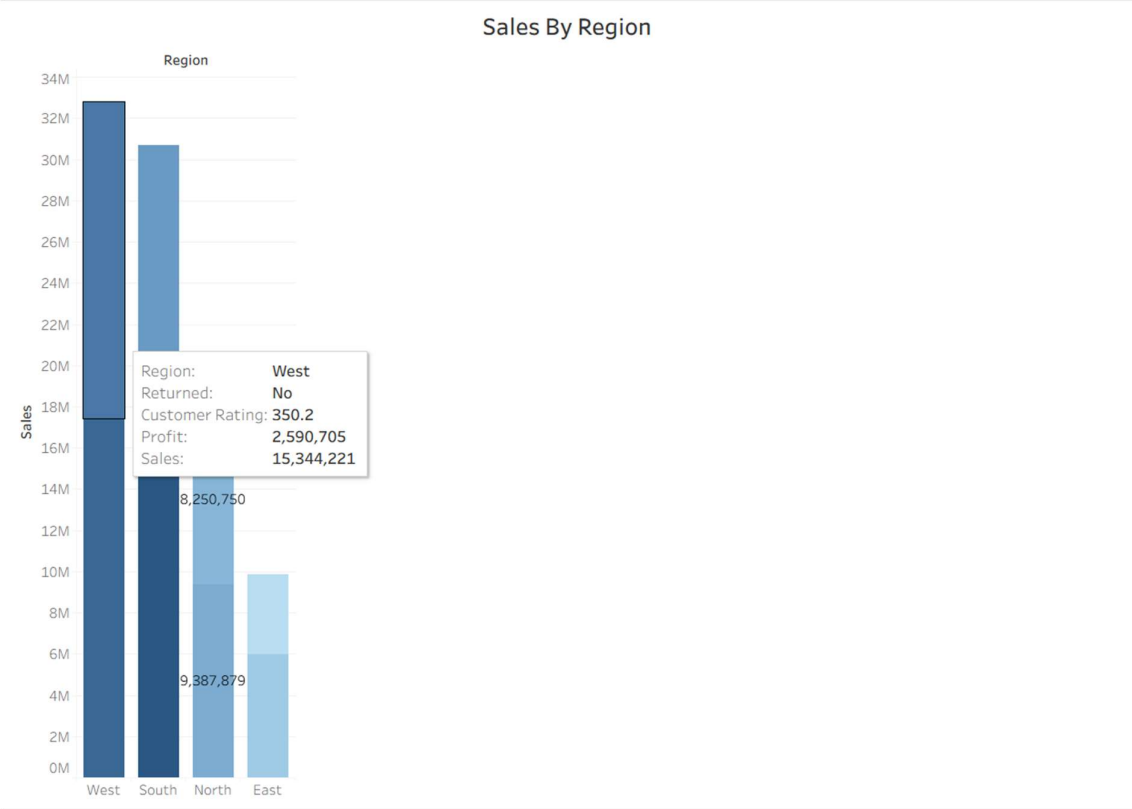
Profit : ₹<SUM(Profit)>

Customer Rating : <AVG(Customer Rating)>

Returned : <Returned>

Expected Output:

When the user hovers over any mark in the visualization, a well-formatted tooltip displays detailed business information related to that data point.



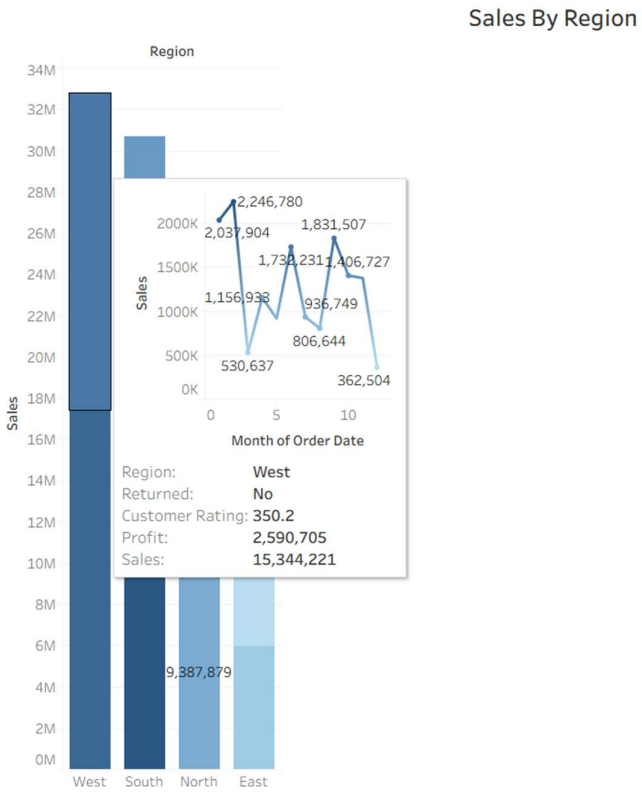
(e) Implement "Viz in Tooltip"

Steps:

1. Create a worksheet named **Monthly Sales Trend** (if not already created).
2. Open the **Sales by Region** worksheet.
3. Click **Tooltip** on the **Marks** card.
4. Click **Insert**.
5. Select **Sheets**.
6. Choose **Monthly Sales Trend**.
7. Click **OK**.
8. Hover the mouse over any region in the chart.

Expected Output:

When the user places the mouse pointer over a region, the tooltip displays a small monthly sales trend chart, providing additional insights without leaving the worksheet.



(f) Improve Readability and User Experience

Steps:

1. Use meaningful titles for all worksheets.
2. Increase font size for titles and labels if necessary.
3. Remove unnecessary grid lines and borders.
4. Keep equal spacing between charts.
5. Add filters such as **Region**, **Product Category**, or **Customer Segment**.
6. Place filters at the top or left side of the dashboard.
7. Display legends in an appropriate location.
8. Ensure charts do not overlap.
9. Maintain a consistent font style throughout the dashboard.

Expected Output:

The dashboard becomes clean, organized, and easy to understand. Users can quickly analyze information and interact with filters without confusion.

(g) Apply Best Practices for Visual Storytelling

Steps:

Step 1: Open the Completed Dashboard

1. Open the dashboard that contains all the worksheets:
    - Sales by Region
    - Monthly Sales Trend
    - Profit by Category
    - Sales by State
    - Customer Rating Distribution

Step 2: Add a Meaningful Dashboard Title

1. Click **Dashboard** from the menu bar.
2. Select **Show Title** (if it is not already enabled).
3. Double-click the dashboard title.
4. Type:

Retail Store Performance Dashboard

5. Change the font to:
    - **Arial**
    - **18 pt**
    - **Bold**
6. Click **OK**.

Purpose: A meaningful title helps users understand the dashboard immediately.

Step 3: Arrange Worksheets in a Logical Sequence

Arrange the worksheets in the following order:

Retail Store Performance Dashboard

Sales by Region Monthly Sales Trend

Profit by Category Sales by State

Customer Rating Distribution

Purpose: Place related charts together so users can understand the information easily.

Step 4: Place Important Charts at the Top

1. Drag the **Sales by Region** worksheet to the upper-left corner.
2. Drag **Monthly Sales Trend** beside it.
3. Place **Profit by Category** below Sales by Region.
4. Place **Sales by State** beside Profit by Category.
5. Place **Customer Rating Distribution** at the bottom.

Purpose: Users usually read from top to bottom, so important information should appear first.

Step 5: Use Consistent Colors

1. Open the **Sales by Region** worksheet.
2. Click **Color** on the **Marks** card.
3. Select a suitable color palette.
4. Repeat the same color theme for all worksheets.
5. Avoid using too many bright or different colors.

Example:

- Blue → Sales

- Green → Profit
- Orange → Ratings

Purpose: Consistent colors improve readability and make the dashboard look professional.

Step 6: Highlight Important Trends

Method 1: Show Labels

1. Open **Sales by Region**.
2. On the **Marks** card, click **Label**.
3. Check **Show Mark Labels**.

Now sales values appear on each bar.

Method 2: Add Annotation

1. Right-click the highest sales bar.
2. Select **Annotate → Mark**.
3. Enter the text:

Highest Sales Region

4. Click **OK**.

The annotation highlights the most important data point.

Step 7: Remove Unnecessary Elements

1. Remove unwanted legends (if not required).
2. Remove unnecessary grid lines:
    - Go to **Format → Lines**.
    - Set Grid Lines to **None**.
3. Remove extra borders if they make the dashboard cluttered.
4. Keep only the charts required for analysis.

Purpose: A clean dashboard is easier to understand.

Step 8: Review the Dashboard

Check the following:

✓ All charts are visible.

✓ Titles are meaningful.

✓ Fonts are readable.

✓ Charts are properly aligned.

✓ Colors are consistent.

✓ Filters are working correctly.

✓ No chart overlaps another.

Step 9: Identify Business Insights

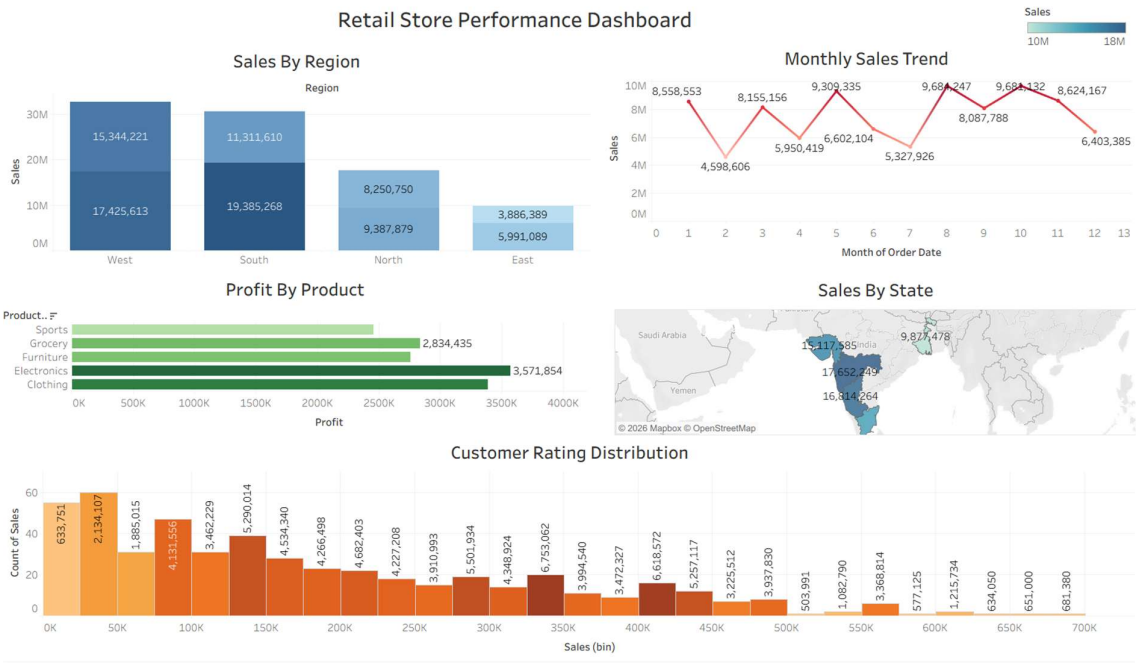
Observe the dashboard and write important findings.

Example Insights

- West Region recorded the highest sales.
- Electronics products generated the maximum profit.
- Customer ratings are generally above 4.5.
- Sales increased during the middle months of the year.
- Returned orders are comparatively low.

Expected Output

A professional dashboard with a clear title, logical chart arrangement, consistent colors, highlighted business insights, and an attractive layout that effectively communicates business performance.



Conclusion:-

This practical demonstrated how Tableau dashboards can be made more professional, interactive, and user-friendly through effective formatting and visualization techniques. Worksheet-level and workbook-level formatting helped maintain consistency, while customized number, date, and null-value formats improved clarity. Appropriate colour schemes and layouts made the dashboards visually appealing and easier to understand. Informative tooltips and Viz in Tooltip provided additional details without overcrowding the dashboard. Overall, applying readability principles and visual storytelling best practices helped transform raw data into meaningful insights that can support effective business decision-making.